

Education

Ph.D. Materials Science and Engineering, <i>University of Wisconsin-Madison</i>	(GPA: 3.70/4.0)	Aug 2024
M.S. Materials Science and Engineering, <i>University of Wisconsin-Madison</i>	(GPA: 3.70/4.0)	Dec 2020
B.S. Engineering, <i>Fort Lewis College</i>	(GPA: 3.99/4.0)	Dec 2017

Skills

Python, PyTorch, scikit-learn, Bash, Git, $\text{\LaTeX}$, C++, OpenHPC, Vim, Docker, Apptainer, Linux, VASP, LAMMPS

Technical Experience

Postdoctoral Appointee **Dec 2024 — Present**
Center for Integrated Nanotechnologies, Sandia National Laboratories Albuquerque, NM

- **Graph Neural Networks:** Built graph neural networks to predict metal-organic framework properties with uncertainty estimates.
- **Bayesian Optimization:** Screened metal-organic frameworks with graph neural network surrogates for favorable separation properties of volatile organic compounds.

Research Assistant **Jun 2018 — Aug 2024**
Computational Materials Group, UW-Madison Madison, WI

- **Machine Learning Domain of Applicability for Materials:** Developed a method using kernel density estimation to assess the applicability domain of machine learning models. The method effectively distinguishes chemically distinct groups and relates high dissimilarity with poor model performance and unreliable uncertainty estimates.
- **Scientific Cluster Construction and Administration:** Assisted in building and managing two high-performance computing clusters. Compiled materials research software and implemented Environment Modules to streamline software management.
- **Quantifying Metallic Glass Forming Ability:** Developed a high-throughput workflow for efficiently training machine learning interatomic potentials and simulating complex material properties. Properties were used to develop models for metallic glass forming ability and followed physically rational trends observed in previous research.

Summer Undergraduate Research Experience **May 2017 — Aug 2017**
Computational Nuclear Engineering Research Group, UW-Madison Madison, WI

- **Developed Python Tool:** Automated the visualization of Direct Accelerated Geometry Monte Carlo (DAGMC) geometries in VisIt.
- **Collaboration Friendly Code:** The Python PEP 8 coding style was adopted to facilitate easier reading.

Capstone Design Project **Sep 2016 — Apr 2017**
Undergraduate Research, Fort Lewis College Durango, CO

- **Built Instrumentation:** Team designed and built an exotic propulsion test stand.
- **MATLAB Modeling:** Implemented electrostatic displacement mechanism and modeled system response.

Summer Undergraduate Research Fellowship **May 2016 — Aug 2016**
Advanced Diagnostics and Propulsion Research Laboratory, Purdue West Lafayette, IN

- **Experimental Imaging:** Operated pressure vessels, X-ray tube sources, and high-speed cameras for analysis of two dimensional sprays.
- **Lead Shielding:** Constructed a protective lead enclosure for X-ray tube sources to shield operating personnel from excessive radiation exposure.

Undergraduate Research, Fort Lewis College Durango, CO

- **Sensor Package:** Designed and developed an interchangeable sensor package for measurement of water temperature, oxygen reduction potential, pH, time, and global positioning system data.
- **Simple Deployment:** Package designed to be durable, waterproof, and easy to use by attaching to rafts or kayaks.

Journal Publications

- Lane E. Schultz, Yiqi Wang, Ryan Jacobs, and Dane Morgan. “A General Approach for Determining Applicability Domain of Machine Learning Models”. In: *npj Computational Materials* (2025). DOI: [10.1038/s41524-025-01573-x](https://doi.org/10.1038/s41524-025-01573-x)
- Lane E. Schultz, Benjamin Afflerbach, Paul M. Voyles, and Dane Morgan. “Machine learning metallic glass critical cooling rates through elemental and molecular simulation based featurization”. In: *Journal of Materiomics* (2024). DOI: [10.1016/j.jmat.2024.100964](https://doi.org/10.1016/j.jmat.2024.100964)
- Lane E. Schultz, Benjamin Afflerbach, Izabela Szlufarska, and Dane Morgan. “Molecular dynamic characteristic temperatures for predicting metallic glass forming ability”. In: *Computational Materials Science* (2022). DOI: [10.1016/j.commatsci.2021.110877](https://doi.org/10.1016/j.commatsci.2021.110877)
- Lane E. Schultz, Benjamin Afflerbach, Carter Francis, Paul M. Voyles, Izabela Szlufarska, and Dane Morgan. “Exploration of characteristic temperature contributions to metallic glass forming ability”. In: *Computational Materials Science* (2021). DOI: [10.1016/j.commatsci.2021.110494](https://doi.org/10.1016/j.commatsci.2021.110494)
- Jun Meng, Md Sariful Sheikh, Lane E. Schultz, William O. Nachlas, Jian Liu, Maciej P. Polak, Ryan Jacobs, and Dane Morgan. “Ultrafast Oxygen Conduction in Sillén Oxychlorides”. In: *Chemistry of Materials* (2025). DOI: doi.org/10.1021/acs.chemmater.4c03163
- Vidit Agrawal, Shixin Zhang, Lane E. Schultz, and Dane Morgan. “Accelerating ensemble uncertainty estimates in supervised materials property regression models”. In: *Computational Materials Science* (2025). DOI: [10.1016/j.commatsci.2024.113494](https://doi.org/10.1016/j.commatsci.2024.113494)
- Ryan Jacobs, Lane E Schultz, Aristana Scourtas, KJ Schmidt, Owen Price-Skelley, Will Engler, Ian Foster, Ben Blaiszik, Paul M Voyles, and Dane Morgan. “Machine learning materials properties with accurate predictions, uncertainty estimates, domain guidance, and persistent online accessibility”. In: *Machine Learning: Science and Technology* (2024). DOI: [10.1088/2632-2153/ad95db](https://doi.org/10.1088/2632-2153/ad95db)
- KJ Schmidt, Aristana Scourtas, Logan Ward, Steve Wangen, Marcus Schwarting, Isaac Darling, Ethan Truelove, Aadit Ambadkar, Ribhav Bose, Zoa Katok, Jingrui Wei, Xiangguo Li, Ryan Jacobs, Lane Schultz, Doyeon Kim, Michael Ferris, Paul M. Voyles, Dane Morgan, Ian Foster, and Ben Blaiszik. “Foundry-ML - Software and Services to Simplify Access to Machine Learning Datasets in Materials Science”. In: *Journal of Open Source Software* (2024). DOI: [10.21105/joss.05467](https://doi.org/10.21105/joss.05467)
- Benjamin T. Afflerbach, Carter Francis, Lane E. Schultz, Janine Spethson, Vanessa Meschke, Elliot Strand, Logan Ward, John H. Perepezko, Dan Thoma, Paul M. Voyles, Izabela Szlufarska, and Dane Morgan. “Machine Learning Prediction of the Critical Cooling Rate for Metallic Glasses from Expanded Datasets and Elemental Features”. In: *Chemistry of Materials* (2022). DOI: [10.1021/acs.chemmater.1c03542](https://doi.org/10.1021/acs.chemmater.1c03542)
- Benjamin T. Afflerbach, Lane Schultz, John H. Perepezko, Paul M. Voyles, Izabela Szlufarska, and Dane Morgan. “Molecular simulation-derived features for machine learning predictions of metal glass forming ability”. In: *Computational Materials Science* (2021). DOI: [10.1016/j.commatsci.2021.110728](https://doi.org/10.1016/j.commatsci.2021.110728)

- J. Xi, G. Bokas, L.E. Schultz, M. Gao, L. Zhao, Y. Shen, J.H. Perepezko, D. Morgan, and I. Szlufarska. “Microalloying effect in ternary Al-Sm-X (X=Ag, Au, Cu) metallic glasses studied by ab initio molecular dynamics”. In: *Computational Materials Science* (2020). doi: [10.1016/j.commatsci.2020.109958](https://doi.org/10.1016/j.commatsci.2020.109958)
- B.R. Halls, J.R. Gord, L.E. Schultz, W.C. Slowman, M.D.A. Lightfoot, S. Roy, and T.R. Meyer. “Quantitative 10-50 kHz X-ray radiography of liquid spray distributions using a rotating-anode tube source”. In: *International Journal of Multiphase Flow* (2018). doi: [10.1016/j.ijmultiphaseflow.2018.07.014](https://doi.org/10.1016/j.ijmultiphaseflow.2018.07.014)

arXiv

- Ryan Jacobs, Maciej P. Polak, Lane E. Schultz, Hamed Mahdavi, Vasant Honavar, and Dane Morgan. “Regression with Large Language Models for Materials and Molecular Property Prediction”. In: *Digital Discovery* (2024). doi: [10.48550/arXiv.2409.06080](https://doi.org/10.48550/arXiv.2409.06080)

Presentations

- Lane Schultz, Benjamin T. Afflerbach, and Dane Morgan. “Molecular Dynamic Characteristic Temperatures for Predicting Metallic Glass Forming Ability”. In: Materials Science & Technology. Columbus, OH, 2021
- Lane Schultz, Benjamin T. Afflerbach, and Dane Morgan. “Molecular Dynamics Features for Predicting Metallic Glass Critical Casting Thickness”. In: Virtual Materials Research Society Spring/Fall Meeting & Exhibit. 2020
- Lane E. Schultz, Thomas J. Cogger, Ryan Good, James Schneider, Robert Rothschild, and William Nollet. “Design of Torsional Test Stand for Micro-Newton Force Detection”. In: 2018 Aerodynamic Measurement Technology and Ground Testing Conference. 2018. doi: [10.2514/6.2018-3737](https://doi.org/10.2514/6.2018-3737)
- James Schneider, Lane E. Schultz, Sophie Mancha, Eric Hicks, and Ryan N. Smith. “Development of a portable water quality sensor for river monitoring from small rafts”. In: OCEANS 2016 MTS/IEEE Monterey. 2016. doi: [10.1109/OCEANS.2016.7761392](https://doi.org/10.1109/OCEANS.2016.7761392)

Teaching Experience

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| • Assisting undergraduates with research | Undergraduate |
| • Assisting laboratory group peers with software installation and cluster use | Graduate |
| • Assisted in class labs (molecular dynamics) | Graduate |
| • Grader for Thermodynamics of Solids | Graduate |
| • Public school substitute teacher | High/Middle/Elementary |
| • Teaching assistant for Thermal and Fluid Systems Laboratory | Undergraduate |
| • Teaching assistant for Engineering Fundamentals II (MATLAB) | Undergraduate |

Awards, Honors, and Societies

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| • PPG Fellowship | University of Wisconsin-Madison, Madison, WI |
| • Ying Yu Chuang Graduate Support Award | University of Wisconsin-Madison, Madison, WI |
| • Sigma Pi Sigma (Physics Honor Society) | Fort Lewis College, Durango, CO |
| • Order of the Engineer | Fort Lewis College, Durango, CO |

Additional Information

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| Citizenship | United States, Colombia, and the Chickasaw Nation |
| Languages | English and Spanish (fluent written and verbal) |